

Fig. 2: An actual-size single-side PCB layout for low-cost audio mixer

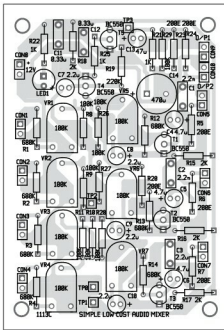


Fig. 3: Component layout for the PCB

resistance in their case is above 2 kilo-ohm, which is an appropriate value for most microphones. The gain of the stages with transistors T1 through T3 is set with the values of resistors R15, R16 and R17, and the gain is usually from 30 to 120—depending on the microphones. The collector currents of transistors T1 through T3 are usually from 0.1mA to 1mA, and are set by resistors R12 through R14, respectively. The collector current of transistor T4 is usually from 0.1mA to 1mA, and for T5 it is from 2mA to 6mA. These currents are set by the resistor R19. The gain of the stage with T4 and T5 is roughly equivalent to the ratio of R18 and R20, and it is set to 10.

CON8 is for connecting 12V power supply. CON9 and CON10 are two output connectors. The power supply is limited mainly by the transistors and amplitude of the signals. The power

supply of the audio mixer is at least 9V. To avoid overloading, it is preferable to use more than 15V. In practice, it is possible to use one or two 9V batteries; alternatively, you can also use two, three or four of 4.5V batteries. One or two 12V rechargeable accumulators also could do the job.

Resistors R1 to R4 can have any value between 10 kilo-ohm and 1 mega-ohm, but this will affect both the input resistance and the sensitivity of the corresponding signal; so there will be some compromises. Resistors R5, R6 and R7 can have any value between 100 ohms and 1 kilo-ohm. Capacitors C1 through C3 should have appropriate values, because they form high-pass filters along with resistors R5 through R7 and input resistances of the microphones. The mixer can be kept on a breadboard until appropriate values of the components are found for par-

ticular configuration of the signal sources and output loads.

The mixer has two identical non-regulated outputs that can drive 600-ohm loads. These can be used to drive both channels of the home stereo system. Also, one of the outputs can drive an audio line and the other can drive high impedance headphones. Both outputs can be paralleled, if necessary. The mixer does not have any expensive components, and does not need complex adjustments, which makes its implementation economical, fast, reliable and with low noise. It can be made even simpler if the potentiometers are omitted.

Construction and testing

The single-side PCB for the simple low-cost audio mixer is shown in actual size in Fig. 2 with its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable case. Fix connectors CON1 through CON7 on front side of the case to connect the seven inputs. Similarly, fix CON8 through CON10 on the rear side to connect the power supply and output to two channels of the audio power amplifier.

To test the circuit, check the input power supply at TP1 with respect to TP0. The circuit is simple and will work immediately after proper assembly. You can check the mixed audio at TP2, and the amplified audio can be observed at TP3 with an oscilloscope. ●

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